

MECH5200 Numerical Methods for PDEs : *Video 19:* *Introduction to Finite Volume Methods*

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References and Acknowledgements

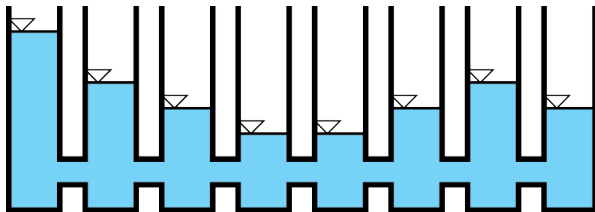
The following materials were used in the preparation of this lecture:

- ① Randall J. LeVesque *Numerical Methods for Conservation Laws*.
- ② 16.920, lecture 11,12 Notes
- ③ Sethian, J.A., *Level Set Methods – Evolving interfaces in Geometry, Fluid Mechanics*
- ④ Versteeg and Malalasekera, *An introduction to computational fluid dynamics – the finite volume method*

The author of these slides wishes to thank these sources for making the current lecture.

Conservation of Mass Example

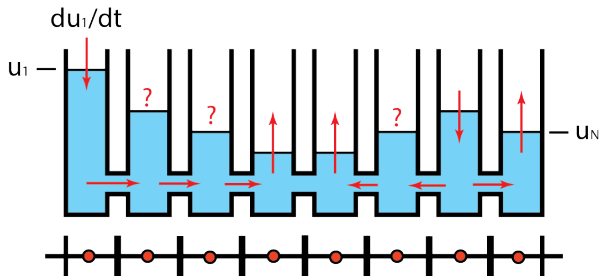
- What if we had a collection of volumes attached by pipes:



- What happens over time?

Conservation of Mass Example

- We know the finite volumes solely gaining/losing fluid will rise/fall respectively:

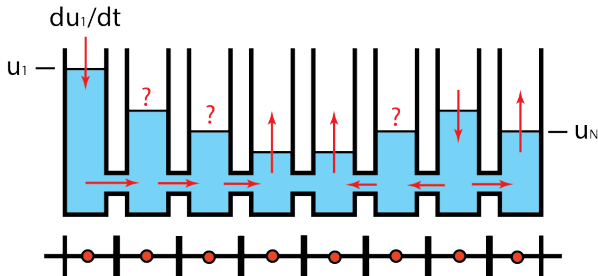


- How can we assess the volumes with inflow and outflow?

$$\frac{du}{dt} = ? \quad (1)$$

Conservation of Mass Example

- We know the finite volumes solely gaining/losing fluid will rise/fall respectively:



- How can we assess the volumes with inflow and outflow?

$$\frac{du}{dt} = flow_{in} - flow_{out} \quad (2)$$

- This is the idea behind finite volumes...

Control Volume Mathematics

- We could recall from fluid dynamics that there is a control volume representation for this situation:

$$0 = \frac{\partial}{\partial t} \iiint_{\Omega} \rho dV + \iint_{\Gamma} \rho \vec{u} \cdot n dS \quad (3)$$

- This expression relates the change in mass inside the volume with the rate at which it is entering/leaving the volume. We can impose the divergence theorem to transform the second integral (boundary integral):

$$0 = \frac{\partial}{\partial t} \iiint_{\Omega} \rho dV + \iiint_{\Omega} \nabla \cdot (\rho \vec{u}) dV \quad (4)$$

- If we then combining the two integrals:

$$0 = \iiint_{\Omega} \left[\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) \right] dV \quad (5)$$

Control Volume Mathematics

- The result is the familiar *conservation of mass* in differential form

$$0 = \frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{u}) \quad (6)$$

- Core Ideas:
 - The differential conservation of mass equation is a PDE
 - The differential conservation of mass equation describes **conservation**
 - Going backwards through the steps, we can derive a **finite volume** representation.

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General Conservation Laws

- In general, many physical phenomena that change with respect to time are conserved.
 - Conservation of mass (ρ is mass per unit volume)
 - Conservation of Momentum ($\rho \vec{u}$ is momentum per unit volume $\rightarrow$ Newtons Law)
 - Conservation of Energy ($\rho u \cdot u^T$ is energy per unit volume $\rightarrow$ Energy equation)
- The underlying conservation form in 1-Dimension is:

$$\frac{\partial u}{\partial t} + \frac{\partial(f(u))}{\partial x} = 0 \quad (7)$$

- Where:
 - u = the unknown property under consideration – this is the desired solution in each cell
 - $f(u)$ = The flux of the quantity between the control volumes

General Conservation Laws

- The basic conservation form:

$$\frac{\partial u}{\partial t} + \frac{\partial(f(u))}{\partial x} = 0 \quad (8)$$

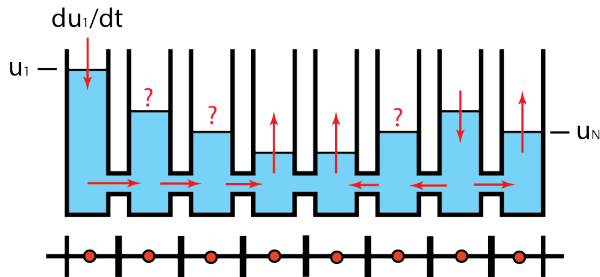
- ASIDE, this can be re-written or expressed as:

$$\frac{\partial u}{\partial t} + \underbrace{\frac{df}{du}}_{a(u)} \frac{\partial u}{\partial x} = 0 \quad (9)$$

- or the (non-linear) advection/wave equation. We saw the linear version ($a(u) = C$) of this last module (hyperbolic PDEs).

Finite Volume Method

- Let's define a 1-D finite volume geometry and discretization:
- We will need to define 1-D "volumes" and "boundaries":



Finite Volume Method

- Our original conservation law in 1-Dimension is:

$$\frac{\partial u}{\partial t} + \frac{\partial(f(u))}{\partial x} = 0 \quad (10)$$

- To begin, we integrate over the domain.

$$0 = \int_{\Omega} \left[\frac{\partial u}{\partial t} + \frac{\partial(f(u))}{\partial x} \right] dV \quad (11)$$

- For our conservation law expression, the integration is the sum of the integrals over the individual elements:

$$\sum_i \int_{x_L}^{x_R} \left[\frac{\partial u_i}{\partial t} + \frac{\partial(f(u_i))}{\partial x} \right] dx = 0 \quad (12)$$

- The conservation law on the i -th individual **line** element:

$$\int_{x_L}^{x_R} \left[\frac{\partial u_i}{\partial t} + \frac{\partial(f(u_i))}{\partial x} \right] dx = 0 \quad (13)$$

Finite Volume Method

- Discussion: By taking this integral, have we changed the nature of the conservation law at all?

Finite Volume Method

- The second integral simply becomes the flux evaluated at the left and right boundaries of the element/cell:

$$\begin{aligned} 0 &= \frac{\partial}{\partial t} \int_{x_L}^{x_R} u dx + \int_{x_L}^{x_R} \frac{\partial f(u)}{\partial x} dx \\ &= \frac{\partial}{\partial t} \int_{x_L}^{x_R} u dx + f(u)_R - f(u)_L \end{aligned}$$

- We can re-arrange this a little as:

$$\frac{\partial}{\partial t} \int_{x_L}^{x_R} u dx = -[f(u)_R - f(u)_L] \quad (14)$$

- This expression says: The change in the property integrated across our control volume is balanced by the flux in and out of the LHS and RHS boundaries of the element.

Finite Volume Method

- Let's focus on the left hand side now.

$$\frac{\partial}{\partial t} \int_{x_L}^{x_R} u dx \quad (15)$$

- We may notice that the average value of the property u across the cell is:

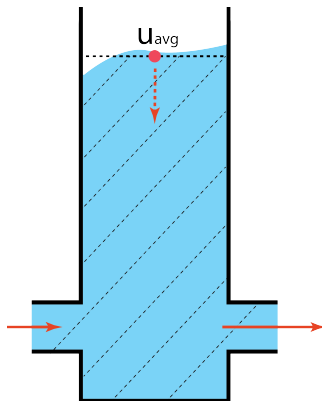
$$\frac{1}{\Delta x} \int_{x_L}^{x_R} u dx = u_{avg} \quad \rightarrow \quad \int_{x_L}^{x_R} u dx = \Delta x u_{avg} \quad (16)$$

- We can re-write our previous expression now as:

$$\frac{\partial}{\partial t} \int_{x_L}^{x_R} u dx = \frac{\partial}{\partial t} (\Delta x u_{avg}) = -[f(u)_R - f(u)_L] \quad (17)$$

Finite Volume Method

- Pictorially, the average value vs. the actual value. What we have effectively done, is assumed a constant value function across each element (most crude approximation we could make, but lets go with it).



Finite Volume Method

- Let's define the time derivative:

$$\frac{\partial}{\partial t}(\Delta x u_{avg}) = -[f(u)_R - f(u)_L] \quad (18)$$

- Discretizing with a forward Euler equation:

$$\frac{(\Delta x u_{avg})^{k+1} - (\Delta x u_{avg})^k}{\Delta t} = -[f(u)_R - f(u)_L] \quad (19)$$

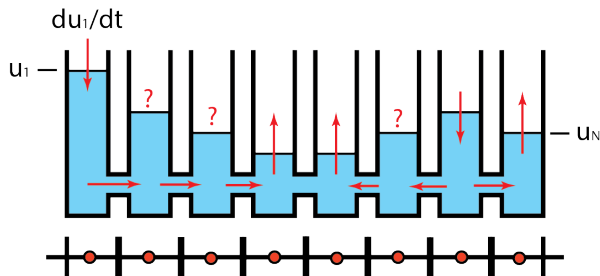
- Rearranging to find the new u :

$$u^{k+1} = u^k - \frac{\Delta t}{\Delta x} [f(u)_R - f(u)_L] \quad (20)$$

- Now we can find the updated u value at some future time.

Finite Volume Method

- Let's Recap:



Finite Volume Method

- Let's try an example. Traffic flow prediction in one-dimension.
- The conservation of cars (mass) equation is:

$$\frac{\partial \rho}{\partial t} + \frac{\partial (f(\rho))}{\partial x} = 0$$

$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u)}{\partial x} = 0$$

- ρ : Density of cars on the road
- u : velocity of the cars

Finite Volume Method

Traffic Flow Example (Leveque):

- We are going to claim there is a relationship between the velocity of the cars and the density:

$$u(\rho) = u_{max} \left(1 - \frac{\rho}{\rho_{max}}\right) \quad (21)$$

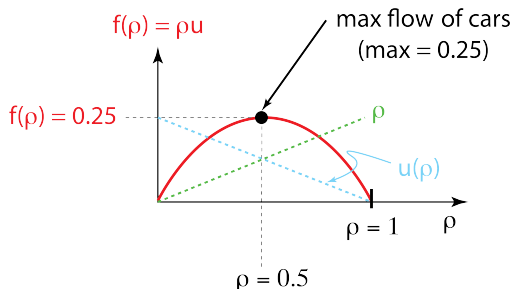
- This equation says that the velocity of cars is faster when $\rho \rightarrow 0$ and slower when $\rho \rightarrow 1$.

Finite Volume Method

- Calculate flow of cars using the velocity-density relationship
- Assume that $u_{max} = 1$ and $\rho_{max} = 1$:

$$f(\rho) = \rho u_{max} \left(1 - \frac{\rho}{\rho_{max}}\right)$$

$$f(\rho) = \rho - \rho^2$$



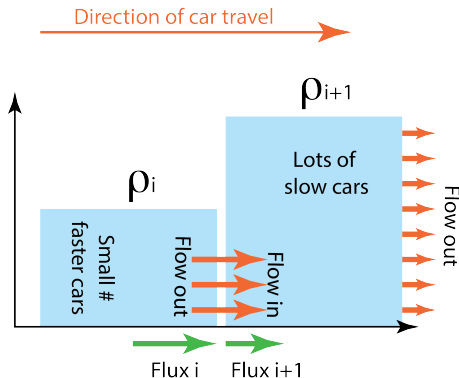
Finite Volume Method

Let's try to implement this for our problem:

- From before, we have:

$$\rho^{k+1} = \rho^k - \frac{\Delta t}{\Delta x} [f(\rho)_R - f(\rho)_L] \quad (22)$$

- Where we need to figure out the $f(\rho)_R$ and $f(\rho)_L$:



Finite Volume Method

- Godunov proposed the following first order flux function:

$$F(\rho_l, \rho_r) = \begin{cases} \min_{(\rho_l \leq \rho \leq \rho_r)} f(\rho) & \text{if } \rho_l \leq \rho_r \\ \max_{(\rho_r \leq \rho \leq \rho_l)} f(\rho) & \text{if } \rho_l > \rho_r \end{cases} \quad (23)$$

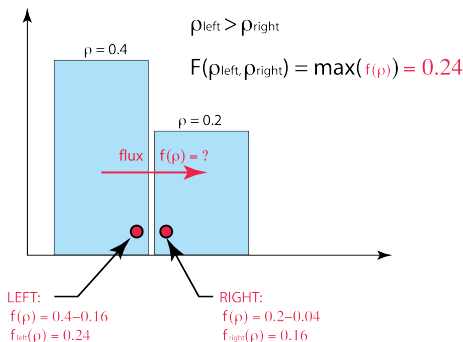
- It basically says, at a boundary between two volumes, you consider the car density/solution values on either side to figure out whether the flow enters/leaves the volume.

Finite Volume Method

- Let's examine a few cases:

$$\rho^{k+1} = \rho^k - \frac{\Delta t}{\Delta x} [f(\rho)_R - f(\rho)_L] \quad \& \quad f(\rho) = \rho - \rho^2 \quad (24)$$

$$F(\rho_l, \rho_r) = \begin{cases} \min_{(\rho_l \leq \rho \leq \rho_r)} f(\rho) & \text{if } \rho_l \leq \rho_r \\ \max_{(\rho_r \leq \rho \leq \rho_l)} f(\rho) & \text{if } \rho_l > \rho_r \end{cases} \quad (25)$$

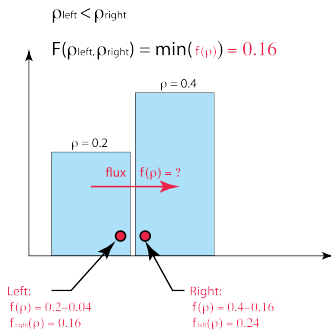


Finite Volume Method

- Let's examine a few cases:

$$\rho^{k+1} = \rho^k - \frac{\Delta t}{\Delta x} [f(\rho)_R - f(\rho)_L] \quad \& \quad f(\rho) = \rho - \rho^2 \quad (26)$$

$$F(\rho_l, \rho_r) = \begin{cases} \min_{(\rho_l \leq \rho \leq \rho_r)} f(\rho) & \text{if } \rho_l \leq \rho_r \\ \max_{(\rho_r \leq \rho \leq \rho_l)} f(\rho) & \text{if } \rho_l > \rho_r \end{cases} \quad (27)$$

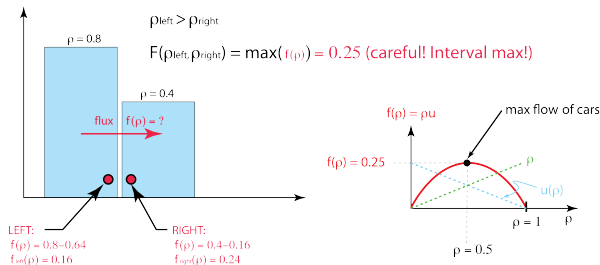


Finite Volume Method

- Let's examine a few cases:

$$\rho^{k+1} = \rho^k - \frac{\Delta t}{\Delta x} [f(\rho)_R - f(\rho)_L] \quad \& \quad f(\rho) = \rho - \rho^2 \quad (28)$$

$$F(\rho_l, \rho_r) = \begin{cases} \min_{(\rho_l \leq \rho \leq \rho_r)} f(\rho) & \text{if } \rho_l \leq \rho_r \\ \max_{(\rho_r \leq \rho \leq \rho_l)} f(\rho) & \text{if } \rho_l > \rho_r \end{cases} \quad (29)$$



Finite Volume Method

- Let's put this all together now:

$$\rho^{k+1} = \rho^k - \frac{\Delta t}{\Delta x} [f(\rho)_R - f(\rho)_L] \quad \& \quad f(\rho) = \rho - \rho^2 \quad (30)$$

$$F(\rho_l, \rho_r) = \begin{cases} \min_{(\rho_l \leq \rho \leq \rho_r)} f(\rho) & \text{if } \rho_l \leq \rho_r \\ \max_{(\rho_r \leq \rho \leq \rho_l)} f(\rho) & \text{if } \rho_l > \rho_r \end{cases} \quad (31)$$

- We know the current density of cars
- We can calculate the flux of cars from one volume to the next (by comparing R and L cells' fluxes and density vals).
- We can then calculate the new value for car density in a block

What have we learned?

- We know that a conservation law can be evaluated as a finite volume method.
- The change of a quantity (ρ) in a volume is related to the flux ($f(\rho)$) of that quantity through the boundary of the cell.
- The flux of the quantity of interest (ρ) through the cell boundaries is determined using a *flux* function.
- Once the fluxes are known, new values of the solution (ρ) can be determined.